Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

1. Aim

To simulate and analyze TEM, TE and TM propagation modes in microwave transmission structures, and to calculate the cutoff frequencies and propagation behavior of a rectangular waveguide operating at 10 GHz.

2. Objectives

- Understand the differences between TEM, TE and TM propagation modes.
- Calculate cutoff frequencies for selected modes of a WR-90 rectangular waveguide.
- Determine whether each mode propagates or is below cutoff at the operating frequency.
- Plot representative transverse field patterns for TE10 and TM11 using MATLAB.
- Relate the calculated results to microwave waveguide operation.

3. Procedure

1.Open MATLAB and create a new script.
2.Enter the operating frequency and waveguide dimensions from the input table.
3.Convert GHz and millimetres to SI units (Hz and metres).
4.Calculate the TE10 and TM11 cutoff frequencies.
5.Compare each cutoff frequency with the operating frequency.
6.Calculate the TE10 propagation constant and guide wavelength.
7.Run the script and inspect the command-window results and field-pattern plots.

4. Theory

For an air-filled rectangular waveguide, the cutoff frequency of TE_mn or TM_mn is fc = (c/2) × sqrt((m/a)^2 + (n/b)^2), where c is the speed of light, a and b are the waveguide width and height, and m and n are mode indices. For TE10, fc = c/(2a). A mode propagates when f > fc and is below cutoff when f < fc.

A hollow, single-conductor rectangular waveguide does not support a TEM mode. TEM propagation is supported by structures such as coaxial cables and parallel-plate lines.

Above cutoff, the phase propagation constant is β = (2πf/c) × sqrt(1 − (fc/f)^2). The formulas assume an air-filled, perfectly conducting waveguide and neglect losses.

5. Input Parameter Table

S. No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
1	Operating frequency	f	10	GHz
2	Waveguide width	a	22.86	mm
3	Waveguide height	b	10.16	mm
4	Waveguide length	L	100	mm

6. MATLAB Code

```
clc; clear; close all;
```

```
a=22.86e-3; b=10.16e-3; c=3e8; f=10e9;
x=linspace(0,a,150); y=linspace(0,b,100); [X,Y]=meshgrid(x,y);
E=exp(-((X-a/2)/(a/4)).^2); H=E/377;
fc1=c/(2*a); fc2=c/2*sqrt((1/a)^2+(1/b)^2); k=2*pi*f/c;
beta1=sqrt(k^2-(2*pi*fc1/c)^2);
x2=linspace(0,a,100); y2=linspace(0,b,100); [X2,Y2]=meshgrid(x2,y2);
E1=sin(pi*X2/a); H1=abs(cos(pi*Y2/b)).^4;
E2=sin(pi*X2/a).*sin(pi*Y2/b); H2=cos(pi*X2/a).*cos(pi*Y2/b);
figure('Color','w');
subplot(2,4,1); contourf(X*1e3,Y*1e3,E,30,'LineColor','none'); colorbar; title('2D TEM Electric Field'); xlabel('X (mm)'); ylabel('Y (mm)');
subplot(2,4,2); contourf(X*1e3,Y*1e3,H,30,'LineColor','none'); colorbar; title('2D TEM Magnetic Field'); xlabel('X (mm)'); ylabel('Y (mm)');
subplot(2,4,3); contourf(X2*1e3,Y2*1e3,E1,20,'LineColor','none'); colorbar; title('TE10 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,4); contourf(X2*1e3,Y2*1e3,H1,20,'LineColor','none'); colorbar; title('TE10 H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,5); contourf(X2*1e3,Y2*1e3,E2,20,'LineColor','none'); colorbar; title('TM11 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,6); contourf(X2*1e3,Y2*1e3,abs(H2),20,'LineColor','none'); colorbar; title('TM11 H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,7); plot([1 2],[fc1 fc2]/1e9,'o-','LineWidth',2); set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); ylabel('GHz'); title('Cutoff Frequency'); grid on;
subplot(2,4,8); plot(1,beta1,'bo','MarkerSize',8,'LineWidth',2); xlim([0.5 2.5]); set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); ylabel('rad/m'); title('Propagation Constant'); grid on;
fprintf('TE10: fc=%.3f GHz, beta=%.2f rad/m\n',fc1/1e9,beta1);
fprintf('TM11: fc=%.3f GHz; below cutoff at 10 GHz.\n',fc2/1e9);
fprintf('TE10 propagates; TM11 does not.\n');
```

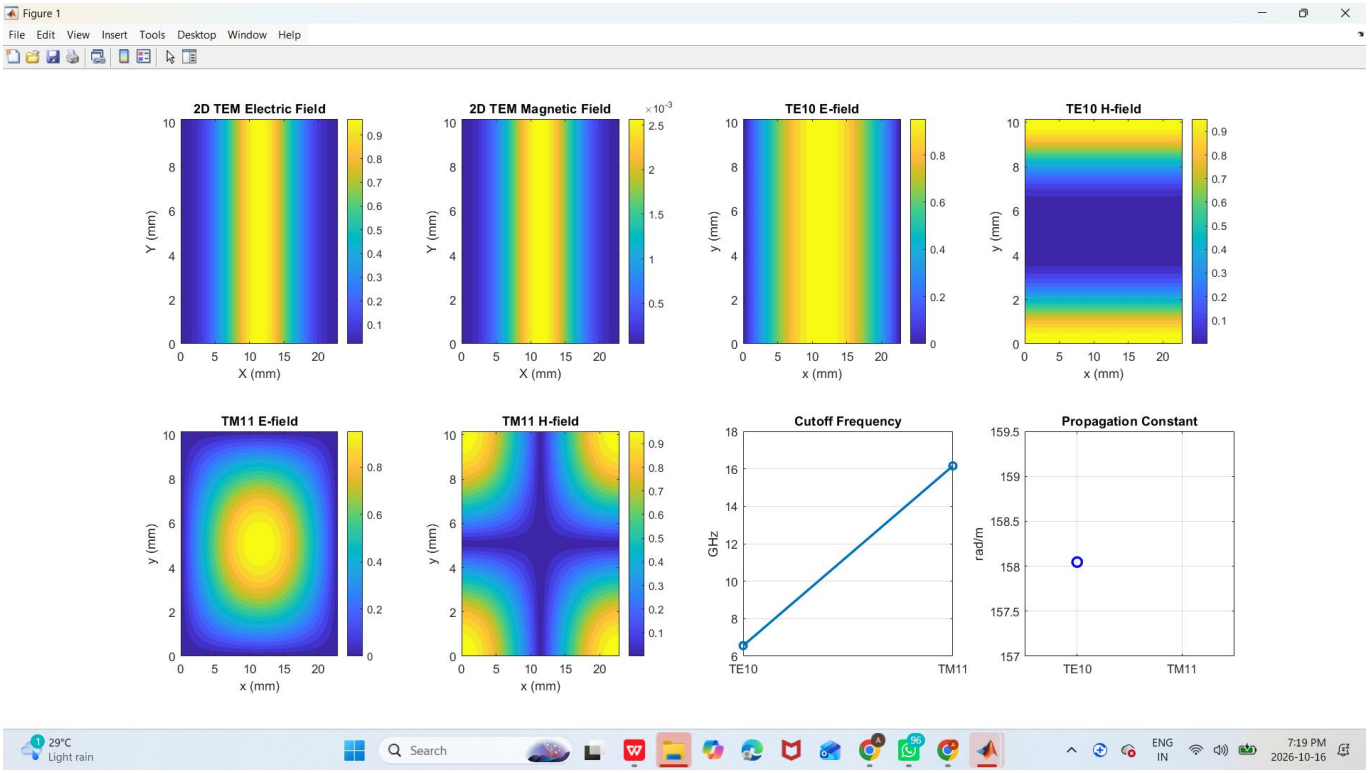
7. Output Parameter Table

S. No.	Output Parameter	Symbol	Calculated Value	Unit / Interpretation
1	TE10 cutoff frequency	fc,TE10	6.562	GHz
2	TM11 cutoff frequency	fc,TM11	16.42	GHz
3	Operating frequency	f	10	GHz
4	TE10 propagation constant	βTE10	≈ 158.0	rad/m
5	TE10 guide wavelength	λg	≈ 39.8	mm
6	TE10 propagation status	f > fc	Propagating	10 GHz > 6.562 GHz
7	TM11 propagation status	f < fc	Below cutoff	10 GHz < 16.42 GHz
8	TEM mode in hollow rectangular waveguide	—	Not supported	Requires suitable multi-conductor structure

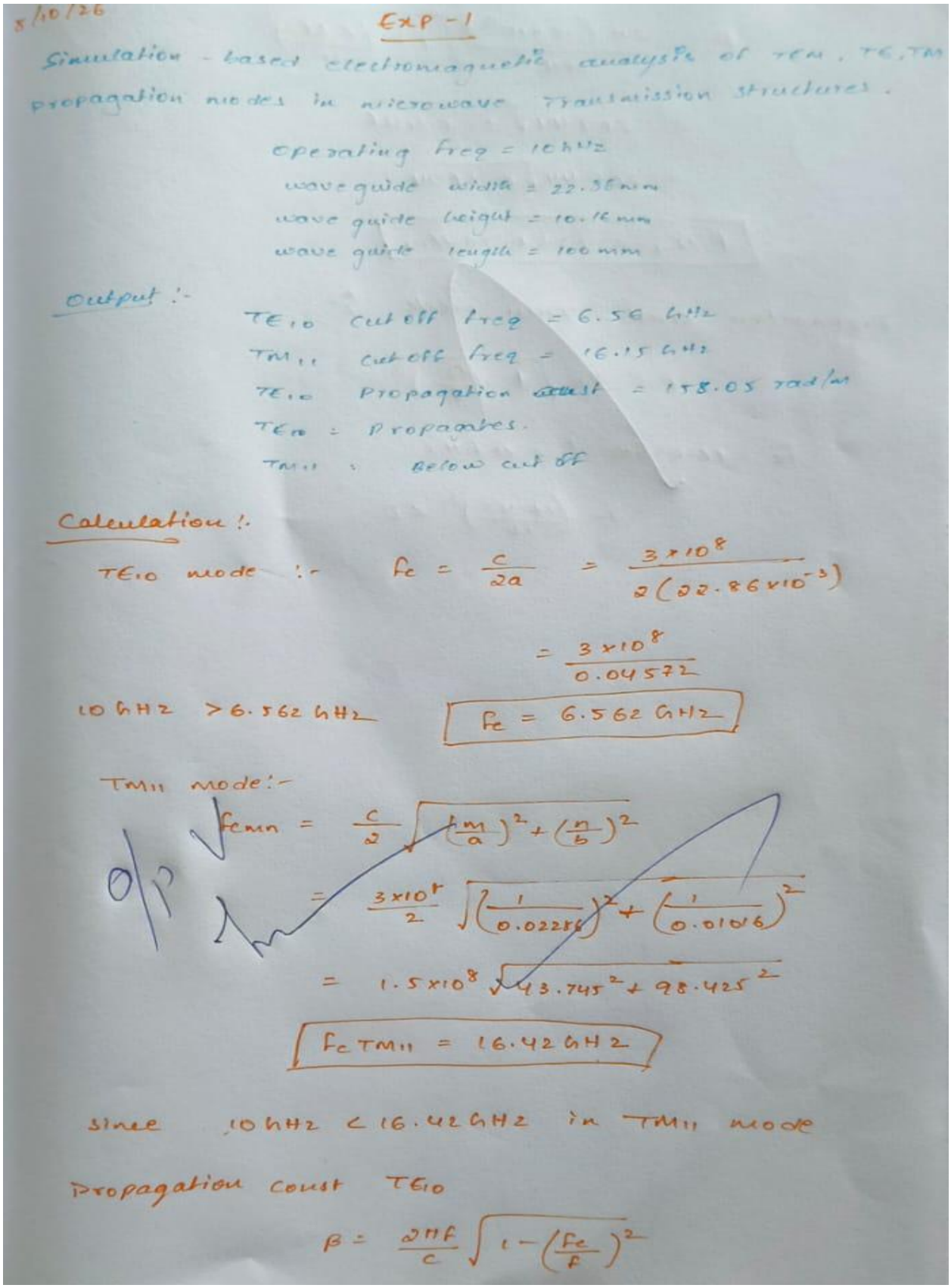
8. Expected Output

The MATLAB command window reports cutoff frequencies, the TE10 propagation constant, guide wavelength and waveguide length. Two contour plots show representative scalar transverse patterns: TE10 varies across the broad dimension, while the TM11 pattern varies in both transverse dimensions. These are illustrative scalar patterns, not full vector electromagnetic field solutions. At 10 GHz, TE10 propagates and TM11 is below cutoff.

OUTPUT/ GRAPH :



Calculations



8/10/26

Exp - 1

Simulation - based electromagnetic analysis of TEM, TE, TM propagation modes in microwave Transmission structures.

Operating freq = 10 GHz
waveguide width = 22.86 mm
wave guide height = 10.16 mm
wave guide length = 100 mm

Output :-

TE_{10} cut off freq = 6.56 GHz
TM_{11} cut off freq = 16.15 GHz
TE_{10} Propagation const = 178.05 rad/m
TE_{10} : Propagates.
TM_{11} : Below cut off

Calculation :-

TE_{10} mode :- $f_c = \frac{c}{2a} = \frac{3\times10^8}{2(22.86\times10^{-3})}$

$= \frac{3\times10^8}{0.04572}$

10 GHz > 6.562 GHz $\quad \boxed{f_c = 6.562\ GHz}$

TM_{11} mode :-

$f_{cmn} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$

$= \frac{3\times10^8}{2}\sqrt{\left(\frac{1}{0.02286}\right)^2+\left(\frac{1}{0.0106}\right)^2}$

$= 1.5\times10^8\sqrt{43.745^2+98.425^2}$

$\boxed{f_c\ TM_{11} = 16.42\ GHz}$

o/p

Since 10 GHz < 16.42 GHz in TM_{11} mode

Propagation const TE_{10}

$\beta = \frac{2\pi f}{c}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$

$$\beta = \frac{2\pi (10\times10^9)}{3\times10^8}\sqrt{1-\left(\frac{6.562}{10}\right)^2}$$

$$= 209.44\sqrt{1-0.4306}$$

$$= 209.44\sqrt{0.5694}$$

$$\boxed{\beta_{TE_{10}} = 158.0 \text{ rad/m}}$$

Propagation const TM11 mode :-

$$\beta = k_0\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

$f_c = 16.42$ GHz $\quad f = 10$ GHz

$$1-\left(\frac{16.42}{10}\right)^2 < 0$$

9. Results

For an air-filled WR-90 rectangular waveguide with a = 22.86 mm and b = 10.16 mm at 10 GHz, the TE10 cutoff frequency is approximately 6.562 GHz, so the dominant TE10 mode propagates. The TM11 cutoff frequency is approximately 16.42 GHz, so TM11 is below cutoff at 10 GHz. The TE10 propagation constant is approximately 158 rad/m and its guide wavelength is approximately 39.8 mm. A TEM mode cannot propagate in a hollow single-conductor rectangular waveguide. These results agree with the expected mode behavior under the stated ideal assumptions.

10. GITHUB LINK

https://github.com/Thanmai1805/ECA1303-Microwave-